

PRATIKRAJ MUGADE

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PROFILE

MSc Data Science (Distinction) graduate with hands-on experience building and deploying production ML systems across healthcare, environmental, life science and enterprise data domains, with hands-on Agentic exposure. Proven ability to clean and analyse large, messy datasets, build and evaluate ML models, and deliver actionable insights to support decision-making. Comfortable across the full stack across data pipelines, model training, FastAPI/LangGraph deployment and CI/CD. Currently working onsite as a GenAI Intern at Virtusa UK.

KEY SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn)

Machine Learning: CNNs, transfer learning, model evaluation, YOLOv11

MLOps & Deployment: AWS Lambda, AWS S3, AWS EC2, Docker, FastAPI, GitHub Actions CI/CD, Git

Data Science (Applied): Classification models, regression models, feature engineering, SMOTE

LLMs & AI Pipelines: LangChain, RAG, OpenAI API, prompt Engineering, vector database, Langfuse

Data Analysis & SQL: SQL (BigQuery, Snowflake), joins, CTEs, window functions, data cleaning, EDA, KPI reporting

Other: Data storytelling, reporting, basic statistics, advanced MS Excel, version control

EXPERIENCE

GenAI Intern – Virtusa UK

Jun '26 – Present

- Architected a multi-user AI research assistant agent (LangChain, LangGraph, Langfuse, ChromaDB, SQLite) with semantic search, ranking, and human-guided curation into cited reports; improved recall@10 nearly 4x via citation-partitioned reranking and hardened the system through RAGAS evaluation and red teaming.
- Co-built a stateful LangGraph platform (Pydantic, FastAPI, PostgreSQL, pgvector, GitHub Actions) that tracks and digests official LLM-Provider announcements and updates; productionised with layered test coverage and Automated CI quality gates.
- Built a chat agent under the mentorship of a senior data architect to reduce rework on existing patterns and optimise SnapLogic credit usage. Developed agent answers the questions over a 500+ pipeline life-sciences data estate with dependency/blast-radius impact scoring and pattern-reuse detection that auto-generates pipeline prompts.

Data Engineering Lead (Co-Founder) – Yabasa Platform

Sep '25 – Jun '26

- Designed data architecture for live peer-to-peer review platform (MySQL, MongoDB, Redis)
- Built automated data validation pipelines to ensure schema integrity and clean data ingestion
- Implemented necessary gateways for supporting safe deployments and data consistency checks
- Supported early product launch and iterative user feedback (200+ active users)

Machine Learning Engineer– Pollinatework Sensibee™ (DEFRA-Regulated Research Contract)

Mar '25 – Aug '25

- Analysed unstructured data generated from IoT field camera and sensor from a DEFRA-regulated ecological monitoring project.
- Built and deployed a series of YOLOv11 computer vision algorithms and automated the git pipelines.
- Cleaned and standardised 10K+ GDPR-compliant records, and translated operational requirements to both technical and non-technical staff working across research, engineering, and commercial teams

Data Science Intern – Healthcare Analytics – Netsoft Technologies, India

May '23 –Dec '23

- Analysed healthcare datasets for chronic kidney disease classification and built a Chronic Kidney Disease diagnostic classifier
- Performed EDA to identify data imbalance and quality issues
- Improved model performance from 95% to 98% using SMOTE and feature refinement

EDUCATION

M.Sc. Data Science (Distinction) from Kingston University, UK

Jan '24 – Jan' 25

B.Eng. Mechanical Engineering (Minor: Computer Science) from Shivaji University, India

Jun '19 – May '23

ACHIEVEMENTS & CERTIFICATIONS

- AWS Certified Cloud Practitioner and AWS Certified AI Practitioner
- Best MSc Project Award from Kingston University

PROJECT WORK

AI Literature Review Agent | FastAPI, LangGraph, React, TypeScript, OpenAI, ChromaDB, SQLite, Docker

- Designed a multi-user research workspace to automate research, ranking and synthesis of academic literature into structured, cited reviews.
- Established rigorous RAG benchmarking using RAGAS, securing a 0.949 faithfulness score across diverse test scenarios, and integrated Langfuse for real-time telemetry and observability.
- Fortified system security by conducting adversarial red-teaming against prompt-injection and cost-abuse vectors, implementing multi-user authentication, usage guardrails, and a hybrid keyword/LLM filtering mechanism.

AI Daily Digest Platform | Python, LangGraph, Pydantic, FastAPI, PostgreSQL, pgvector, GitHub Actions

- Developed a stateful agentic system to ingest official AI company announcements and updates, version historical snapshots, and track automated delta changes over time.
- Orchestrated a custom LangGraph pipeline for structured extraction, comparison, and citation checking, featuring automated human-in-the-loop validation gates for unsupported claims.
- Instituted rigorous engineering standards, including inner/outer loop testing, typed API contracts, ADR documentation, and automated CI quality gates with reasonable coverage.
- Architected a delivery infrastructure combining FastAPI and scheduled email briefs, using PostgreSQL for core data and pgvector for semantic search.

SnapLogic Estate Assistant | AI Agent for Data Pipeline Intelligence under mentorship of a senior data architect

- Co-designed and developed a chat-based AI agent within Databricks to provide intelligent operational insights across a life-sciences estate of 500+ data pipelines.
- Engineered a dependency graph and blast-radius impact analysis, including a composite risk-scoring model (failures × downstream dependents) to map pipeline reliability risk
- Built pattern-matching logic to automatically discover reusable pipeline templates and generate contextual system prompts for rapid, code-driven pipeline creation.
- Presented the working prototype to platform stakeholders as part of a client-facing AI acceleration initiative.

Multi-class Insect Classifier | CNN, AWS EC2, Transfer learning, Computer vision

Aug '25 - Sept '25

- Created a GDPR compliant dataset from scratch using multiple sources.
- Balanced the dataset using a scientific and systematic approach of selective augmentation.
- Optimised the models using hyperparameter tuning and deployed them on EC2 for real-time use cases.
- Recorded an accuracy of 98.99% on the field data.

Data Quality & Analytics Pipeline (Academic Project) | Python, SQL

Oct '24 - Dec '24

- Cleaned and standardised large datasets using Python and SQL.
- Performed structured data validation and quality checks.
- Improved dataset reliability for downstream reporting and analysis.